The American University in Cairo
School of Business
Access to Knowledge for Development Center

A STUDY OF DATA AND ARTIFICIAL INTELLIGENCE NARRATIVES IN MENA

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Abstract

Analyzing context is critical for understanding current capacities in the region to utilize artificial intelligence (AI) technologies and address local challenges as they arise. This study aims to explore the state of AI, in the public and private sectors in select MENA countries. Through semi-structured interviews with stakeholders in 6 MENA countries, the study explores the existing obstacles for AI adoption and production, in addition to recommendations to foster the AI environment. The countries covered are Egypt, Tunisia, Palestine, Jordan, Lebanon, and the United Arab Emirates.

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I. Introduction

Progress in data collection, algorithms and aggregation has allowed scientists and engineers to make great advances in developing artificial intelligence (AI) worldwide. Performing tasks that required human cognition can now be performed by machines.[1] Machine learning systems can adapt to new data without being explicitly reprogramed.[2] These systems are now put to commercial use in many countries. The objective of this research is to gain insights into perspectives on data and artificial intelligence (AI), especially as they pertain to development in select countries of the Middle East and North Africa (MENA) region. This will take place through a series of semi-structured interviews with stakeholders across the ecosystem in seven countries in the region: Egypt, Tunisia, the United Arab Emirates (UAE), Jordan, Lebanon, and Palestine.

The research probes areas including the state of AI in MENA, in terms of adoption and creation, and the relationship between data and artificial intelligence in the MENA region. Stakeholder perspectives on necessary capacities for harnessing AI for development are also investigated, as well as identification of countries' most pressing development needs and what role (if any) AI and new technologies can play in that regard.

II. Overview of AI and Data in MENA: local capacities, know-how, and regulation

In the MENA region, AI adoption is growing in sectors such as construction, manufacturing, energy utility and resources, health care, education, financial services, consumer goods, transport, and telecommunications.[3] Despite the great potential for direct gains in these sectors, data, algorithms and infrastructure arise as points of concern for the MENA region. As noted by Nagla Rizk in her recent chapter "Artificial Intelligence and Inequality in the Middle East: The Political Economy of Inclusion*", in respect of inclusion and sustainable development, "biases in data, black boxes in algorithms, and the inaccessibility and inadequacy of infrastructure can all serve to exclude and marginalize".[4]

A predominantly young population as well as inequities in connectivity across the region and within countries are also of particular importance in MENA. More than 60% of the population is under 25 years, making the region one of the most youthful in the world.[5] While

[1] Darell M. West and John R. Allen, "How artificial intelligence is transforming the world," Brookings, April 2018 https://www.brookings.edu/research/how-artificial-intelligence-is-transforming-the-world/

[2] ● Blog, Machine Learning, "What is Machine Learning? A definition," Expert System, May 2020 https://expertsystem.com/machine-learning-definition/

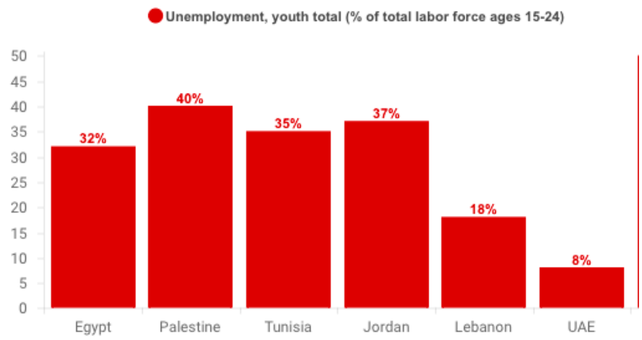
[3] Shivangi Jain, "The potential impact of Artificial Intelligence in the Middle East", *PwC Middle East,* Report Published February 2018. https://www.pwc.com/m1/en/publications/documents/economic-potential-ai-middle-east.pdf

[4] Nagla Rizk, "Artificial Intelligence and Inequality in the Middle East: The Political Economy of Inclusion," In Oxford Handbook of Ethics of Artificial Intelligence, M Dubber, ed. (Oxford: Oxford University Press, 2020).

[5] Yamout Salam, 'Internet Access and Education: Transforming lives in the Middle East', *Internet Society,* August 2018 https://www.internetsociety.org/blog/2018/08/internet-access-and-education-transforming-lives-in-the-middle-east/

this holds great promise, it also raises concerns of unemployment: youth unemployment rates in the MENA region are higher than the world average of 13 percent.[6] Recent estimates (2019) indicate diverse high rates in Palestine (40 percent)[7], Jordan (37 percent), Egypt (32 percent), Lebanon (18 percent) and Tunisia (35 percent).[8] Among this large youthful population, internet access and the development of digital skills (ICT education) are two key factors for integration into the digital economy, which encompasses AI.

Figure 1[9]



Internet access in the GCC (Gulf Cooperation Council) is the highest amongst MENA countries. The internet penetration rate for the UAE is 98.8 percent for 2020[10] and Saudi Arabia 91.4 percent in the same year.[11] In contrast, in North Africa, Egypt has an internet penetration rate of 54 percent.[12] In late 2019 the rate in Tunisia was 66.8 percent.[13] Despite these relatively high internet penetration rates, some countries in the MENA region still face

[6] Nader Kabbani, "Youth employment in the Middle East and North Africa: Revisiting and refraining the challenge," Brookings, February 2019 https://www.brookings.edu/research/youth-employment-in-the-middle-east-and-north-africa-revisiting-and-reframing-the-challenge/

[7] Trading Economics, *Palestine Youth Unemployment Rate 2009-2019,* https://tradingeconomics.com/palestine/youth-unemployment-rate

[8] • World Bank, "Unemployment, youth total (% of total labor force ages 15-24) (modeled ILO estimate) – Saudi Arabia, Jordan, Egypt, Arab Rep, and Tunisia International Labour Organization," ILOSTAT database. Data retrieved in December 2019. https://data.worldbank.org/indicator/SL.UEM.1524.ZS?locations=EG

[9] Data retrieved from: World Bank, *Unemployment, youth total (% of total labor force ages 15-24)* (modeled ILO estimate) – Saudi Arabia, Jordan, Egypt, Arab Rep, Lebanon, Palestine, and Tunisia International Labour Organization, ILOSTAT database. Data retrieved in December 2019. https://data.worldbank.org/indicator/SL.UEM.1524.ZS?locations=EG, Unemployment, youth total (% of total labor force ages 15-24) - (modeled ILO estimate) United Arab Emirates https://data.worldbank.org/indicator/SL.UEM.1524.ZS?locations=AE

[10] GMI Blogger, "UAE Internet & Mobile Statistics 2020 (Infographics)," Global Media Insight, March 2020, https://www.globalmediainsight.com/blog/uae-internet-and-social-media-usage-statistics/

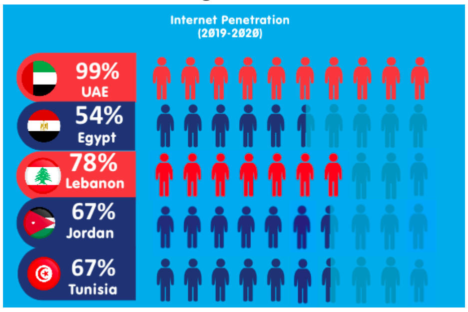
[11] Statistical Research Department, "Saudi Arabia: internet user penetration 2017-2023," Statista, August 2020, https://www.statista.com/statistics/484930/internet-user-reach-saudi-arabia/

[12] Amwal Al Ghad English, "Egypt's internet penetration hits 54% - We are Social," Amwal Al Ghad, February 2020 https://en.amwalalghad.com/egypts-internet-penetration-hits-54-we-are-social/

[13] "Internet World Statistics: Africa," Internet World Stats: Usage and Population Statistics, accessed October 25th, 2020, https://www.internetworldstats.com/africa.htm#ng

issues with quality internet access that is affordable to the majority of the population. Affordable and accessible internet is a necessity to develop digital skills and integrate in the digital economy.

Figure 2[14]



To unlock the potential of AI applications, MENA countries must operate with a relatively high degree of openness. Openness in AI development can refer to "open source code, open science, open data, or openness about safety techniques, capabilities, and organizational goals, or a non-proprietary development regime generally".[15] According to Bostrom, most leading AI developers, and researchers, at Google, Facebook, and Microsoft, operate with a considerable degree of openness.[16] Many MENA AI executives in the private sector are investing in creating an open data architecture for quality data.[17]

Open data can be defined as data that is openly and freely accessible in machine-readable formats for anyone to use, reuse and share.[18] Data is seen as a necessity for AI performance.[19]

[14] Data retrieved from: UAE Internet & Mobile Statistics 2020 (Infographics) https://www.globalmediainsight.com/blog/uae-internet-and-social-media-usage-statistics/, Amwal Al Ghad 'Egypt's internet penetration hits 54% - We are Social' February 2020, https://en.amwalalghad.com/egypts-internet-penetration-hits-54-we-are-social/, Internet World Statistics: Africa https://www.internetworldstats.com/africa.htm#ng), Jordan Internet penetration source: https://datareportal.com/reports/digital-2020-jordan
Lebanon Internet penetration source: https://datareportal.com/reports/digital-2020-lebanon

[15] Nick Bostrom, "Strategic Implications of Openness in AI Development", *Future of Humanity Institute, University of Oxford,* May 2017. https://www.nickbostrom.com/papers/openness.pdf

[16] Nick Bostrom, "Strategic Implications of Openness in AI Development", *Future of Humanity Institute, University of Oxford,* May 2017. https://www.nickbostrom.com/papers/openness.pdf

[17] Thoma Holm Moller, James Matcher, and Ellen Czaika, "Artificial Intelligence in Middle East and Africa: Outlook for 2019 and Beyond", Prepared by EY Consulting LLC for Microsoft, https://info.microsoft.com/rs/157-GQE-382/images/report-SRGCM1065.pdf

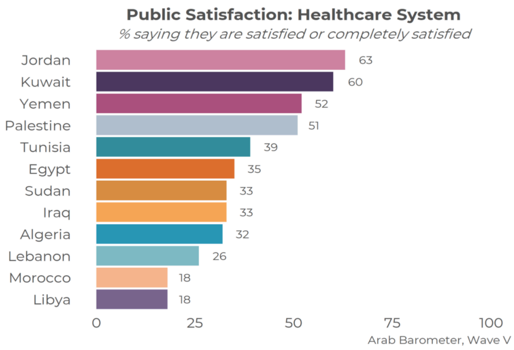
[18] Rayna Stamboliyska, "Open Data Index 2013: Fundamental public sector data still unavailable in MENA". Open Knowledge Foundation Egypt, October 2013, accessed November 26th 2020, https://eg.okfn.org/2013/10/open-data-index-2013-fundamental-public-sector-data-still-inavailable-in-mena/

[19] Publications Office at the European Union, "AI and Open Data: a crucial combination" European Data Portal, April 2018 https://www.europeandataportal.eu/en/highlights/ai-and-open-data-crucial-combination

To function properly, algorithms in AI systems need large quantities of high-quality data. As such, governments and public organizations need to make data openly available. Legislation of open data in MENA can promote and increase access to high-quality data volumes that will enable organizations to unleash the potential of AI, gain reliable insights and allow innovation to flourish. Estimates show that the Middle East is expected to accumulate 2% (equivalent to US $320 billion) of the global benefits of AI in 2030.[20] Perhaps owing to this, the United Arab Emirates, Saudi Arabia, Qatar, Egypt, and Tunisia are already taking bold steps with AI initiatives. These steps include increased investment across sectors, policy awareness and commitment. Across the Middle East, governments and businesses are starting to confront the choice between shifting towards AI and advanced technologies or being left behind with technological disruption.

The COVID-19 pandemic has accelerated the digital transformation globally and in the MENA region specifically. With over 400,000 confirmed cases across the MENA region, the pandemic threatens to overwhelm already overburdened health systems, and further expose existing inequalities in access to health benefits. Furthermore, the pandemic has also highlighted existing health data issues as the region moves towards integrating local and regional health surveillance systems in order to inform evidence-based decisionmaking in containing the outbreak.

Figure 3[21]



On the other hand, the pandemic has also stirred innovation, especially within new and existing start-ups, who are pushing boundaries with new solutions to fill existing and quickly-arising market gaps. Rapid increase in "healthtech" investments and innovators leveraging artificial intelligence and machine learning to enable products and solutions is already underway. The pandemic has also facilitated digital transformation as more and more startups are now offering remote monitoring and telehealth platforms. These new digital platforms are seeing substantial traffic since the outbreak, particularly for their use of AI-enabled assessment apps and devices.

[20]Shivangi Jain, "The potential impact of Artificial Intelligence in the Middle East", *PwC Middle East,* Report Published February 2018.
https://www.pwc.com/m1/en/publications/documents/economic-potential-ai-middle-east.pdf
[21] Erin Hayes, "The Impact of COVID-19 on MENA's Already Challenged Healthcare Systems", Arab Barometer, July 2020,
https://www.arabbarometer.org/2020/07/low-capacity-and-trust-the-impact-of-covid-19-on-menas-already-challenged-healthcare-systems/

The Egyptian government has adopted a policy of increased access to healthcare and fostering cross-industry partnerships. "Digitized hospitals" are fast-becoming a norm in MENA markets, emphasizing connectivity, hardware and software developments related to big data, machine learning, artificial intelligence and the Internet of Things—a pathway for innovators in healthcare worldwide. One example of such innovations is the One4All initiative, essentially a collaboration between existing industry leaders in the healthcare sector and emerging start-ups in the field that aims to provide several online/remote healthcare services to Egyptians free of charge. According to news reports, "the initiative provides on-demand nursing services, free online counselling, including an algorithm-powered chatbot that assesses symptoms to determine the likelihood of them having COVID-19 and need testing, as well as triage sessions."[22] All of these services are offered in both Arabic and English and only require an internet connection.

Another fast-growing digital platform in the healthcare market has been Vezeeta, the leading digital healthcare booking platform operating in Egypt, Saudi Arabia, the Levant and the UAE, and practice management software in MENA that provides healthcare services including medical consultations and booking doctors. According to CEO and co-founder of the UAE-based Vezeeta Amir Barsoum, digital solutions like telemedicine are quickly shifting from a previously slow adoption path to a pace of uptake.[23] Health-tech companies that have primarily been operating at the fringes of the system are stepping up their game to fill existing gaps and reduce the burden on frontline workers and healthcare systems. Barsoum said that since the COVID-19 outbreak in March, Vezeeta launched tele-health in Egypt, Saudi, Jordan, Kenya, and soon Nigeria, generating more than 30,000 calls in the span of a month.[24]

III. What capacities are necessary for AI development?

A number of recent, key articles and reports are assessed in this literature review in an attempt to answer the following overarching question: *what capacities are necessary for Artificial Intelligence (AI) development?* In her chapter, "Artificial Intelligence and Inequality in the Middle East: The Political Economy of Inclusion", Rizk identifies data, infrastructure, human resources and an enabling environment as the key capacities vital to an inclusive development of AI technologies which does not exacerbate existing inequalities in the MENA region. This goes in line with identifications in global literature on AI development, while taking into account the specifics of inequality and possibilities for ethical AI development in the MENA region.

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[22] Doaa A. Moneim, "Healthcare business to witness quantum leap in MENA region amid COVID-19 crisis," Ahram Online, May 2020, http://english.ahram.org.eg/NewsContent/3/12/369718/Business/Economy/Healthcare-business-to-witness-quantum-leap-in-MEN.aspx

[23] Doaa A. Moneim, "Healthcare business to witness quantum leap in MENA region amid COVID-19 crisis," Ahram Online, May 2020, http://english.ahram.org.eg/NewsContent/3/12/369718/Business/Economy/Healthcare-business-to-witness-quantum-leap-in-MEN.aspx

[24] Doaa A. Moneim, "Healthcare business to witness quantum leap in MENA region amid COVID-19 crisis," Ahram Online, May 2020, http://english.ahram.org.eg/NewsContent/3/12/369718/Business/Economy/Healthcare-business-to-witness-quantum-leap-in-MEN.aspx

A. Data

As briefly mentioned earlier, data is the most important capacity for the development of AI identified across the literature. It feeds machine learning and artificial intelligence models and enables them to better perform tasks and respond to challenges. Moreover, data availability and access are necessary for the development of other digital technologies that are in turn necessary for the development of AI. For example, Joshua Meltzer warns against data localization measures, or restrictions on cross-border data flows. He argues that such restrictions will most directly and disproportionately impact developing countries: "to develop AI in areas such as healthcare, countries with smaller populations will require access to global health data. Limits on access to such data will reduce the accuracy and relevance of AI systems for developing countries".[25] He therefore calls for governments with existing repositories of large data sets to enable global access and highlights data trade agreements such as the Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP) and the United States-Mexico-Canada Agreement (USMCA) as examples of policy that will enable ethical AI development.

One possible solution suggested by the literature is the creation of data commons: accessible, interoperable infrastructure that co-locates data, storage, and computes with common analysis tools for the goal of overcoming the present division between data users and producers.[26] Benjamin Kumpf stresses the need for establishing ethical frameworks and accountability measures, such as measuring AI for responsibility, fairness, and legal discrimination, in order to ensure ethical consumption of open data and by extension ethical application of AI.[27] In the case of the MENA region, the Economic Research Forum argues that "many economies in the MENA region have either lagged in their capacity to generate data or they have prevented the research community, the independent media and civil society from accessing the data. While concerns over privacy are real, little attention has been paid to the costs of opaque data systems that hamper external knowledge generation".[28] The World Bank also warns about the risk of "data deprivation" in the Levant, "Lebanon, the Palestinian Territories, Jordan, Syria and Iraq, where conflict and displacement have been particularly acute, no poverty estimates using household surveys have been released since 2012."[29]

Indeed, a more pressing issue concerning data in the MENA region is quality and availability. Rizk encapsulates issues pertaining to data quality in the MENA region under the categories of data asymmetry, "data lock", and data inaccuracy as impediments which must be

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[25] Joshua P. Meltzer, "The Impact of Artificial Intelligence on International Trade," Brookings, accessed April 29th, 2020, https://www.brookings.edu/research/the-impact-of-artificial-intelligence-on-international-trade/

[26] Benjamin Kumpf, 'Who is writing the future? Designing infrastructure for ethical AI," Medium, accessed April 29th, 2020, https://medium.com/@UNDP/who-is-writing-the-future-designing-infrastructure-for-ethical-ai-4999620db295)

[27] Benjamin Kumpf, 'Who is writing the future? Designing infrastructure for ethical AI," Medium, accessed April 29th, 2020, https://medium.com/@UNDP/who-is-writing-the-future-designing-infrastructure-for-ethical-ai-4999620db295)

[28] Asif Islam, "Data Capacity and Transparency in MENA: Why They Might Matter For Growth," The Forum: Economic Research Forum Policy Portal, May 2020. https://theforum.erf.org.eg/2020/05/10/data-capacity-transparency-mena-might-matter-growth/

[29] Aziz Atamanov and Nandini Krishnan, "Improving data collection to improve welfare in the Middle East," World Bank Blogs, October 2016. https://blogs.worldbank.org/arabvoices/improving-data-collection-improve-welfare-middle-east

overcome to realize inclusive AI development.[30] Data asymmetry and data lock both refer to restrictions on access to data by individuals or small enterprises, the first due to a disparity between large companies who can afford to gather or solicit large data sets and smaller companies who cannot, and the latter due to restrictions on access to national statistics data by the state.[31] Indeed, a more pressing issue concerning data in the MENA region is quality and availability. The Open Data Barometer reports that only about 66% of data considered for analysis in the region was available online, and points to the fact that, in the case of Egypt for example, "datasets are frequently outdated and face licensing, quality, and discoverability issues, often making them useless".[32] However, countries in the GCC began implementing a number of big data initiatives to tackle these issues in recent years, such as the UAE's Smart Data Strategy, or the Qatar Smart Program (TASMU), Bahrain Open Data and KSA Open and Big Data strategies.[33]

B. Enabling environment

Another prerequisite for inclusive AI development is an enabling environment.[34] This capacity is articulated in different terms across the literature, but it all points to the leading role of governments in creating this environment as well as harnessing the potential of the private sector and guiding it towards ethical AI development. Tanya Filer, writing for the Bennet Institute of Public Policy, argues for the need to integrate AI capabilities within government.[35] For example, AI researchers should maintain a presence within government departments. It is also recommended that policymakers and senior officials already in the government receive AI training. The second suggested policy by Filer relevant to the question of capacities required for AI includes that a push be made by the government towards encouraging its investments to AI-startups in the direction of developing and enhancing core public services, such as social welfare. This can be done through a careful redirection and reassessment of research and development funding by the state. In her recent chapter, Rizk identifies an enabling legislative environment in which there exists an "ecosystem of legislation that supports innovation, access to markets, open data, and building human capacity and technology development".

Furthermore, the scholarship points to the role of government in implementing comprehensive strategies in order to manage socio-economic disruptions caused by the introduction of AI technology. This can be achieved by ensuring that AI reflects national values and interests – a reasonable threat given that most AI development takes place in a

[30] Nagla Rizk, "Artificial Intelligence and Inequality in the Middle East: The Political Economy of Inclusion," In Oxford Handbook of Ethics of Artificial Intelligence, M Dubber, ed. (Oxford: Oxford University Press, 2020).
[31] Nagla Rizk, "Artificial Intelligence and Inequality in the Middle East: The Political Economy of Inclusion," In Oxford Handbook of Ethics of Artificial Intelligence, M Dubber, ed. (Oxford: Oxford University Press, 2020).
[32] "Regional Snapshot: Middle East and North Africa Country Profiles," Open Data Barometer, https://opendatabarometer.org/4thedition/regional-snapshot/middle-east-north-africa/#country-profiles
[33] Bhavesh Morar, Maher Kilani, Jean Louis Prevost, and Kristian Meyer, "Big Data in the GCC: From Strategy to Global Leadership". Deloitte, ME PoV Summer 2019 Issue, accessed November 26th, 2020, https://www2.deloitte.com/ye/en/pages/about-deloitte/articles/revolution/big-data-gcc.html
[34] Nagla Rizk, "Artificial Intelligence and Inequality in the Middle East: The Political Economy of Inclusion," In Oxford Handbook of Ethics of Artificial Intelligence, M Dubber, ed. (Oxford: Oxford University Press, 2020).
[35] Tanya Filer, "Developing AI for Government: What role and limits for the private sector?" Bennet Institute, accessed April 29th, 2020, https://www.bennettinstitute.cam.ac.uk/blog/developing-ai-government-what-role-and-limits-priv/

concentrated area within the Global North (Silicon Valley). Among the approaches are the instrumentalization of AI for addressing regional skill gaps, the improvement of decision-making processes and enhancing cybersecurity capabilities.[36] Richard Stirling and Hannah Miller suggest reframing the discussion around AI in terms of the question, what can the government do with the help of AI, to "deliver better public services, to more people, more cheaply?"[37] The authors then turn to suggest ways to increase national AI capacities through government regulation of AI technology development activities such as primary research, applied research, and the implementation of a framework for artificial general intelligence all centered around national human interests.

C. Infrastructure

Adequate infrastructure is identified as a further necessary aspect of equitable and inclusive AI development. Rizk specifically highlights the "uneven access to data storage and computing capacity, limited internet connectivity and host of issues related to how algorithms are intertwined with the human context".[38] This description captures the most pressing infrastructure issues impeding the development of ethical AI in the MENA region. The literature recommends enhancement and building upon existing infrastructure as necessary capacities for AI development. Talton and Tonar suggest ensuring that existing expertise and experience made available by the current public and private sector infrastructure leaders be "augmented by perspectives from computer science and software engineering", since such expertise, as it exists today, remains largely influenced by mechanical, civil, construction, electrical and other specializations outside the AI sphere of knowledge.[39] Such new perspectives can provide the necessary support for existing traditional infrastructural knowledge with a much-needed digital intelligence perspective.

D. Human resources

Additionally, human resources are underscored as a significant enabler for AI development. This capacity, as identified in the literature, encompasses different but complementary understandings of the term. James Manyika and Jacques Bughin, researchers at the Mckinsey insititute, predict that the ability to innovate and acquire human capital skills will prove to be one of the most important enablers for AI development.[40] Rizk also points to human capital as an invaluable resource for but also one which is threatened by the emergence of AI technologies in the MENA context. This is echoed by another report on the economic

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[36] Jeff Loucks, Susanne Hupfer, David Jarvis and Timothy Murphy, "Future in the balance? How countries are pursuing an AI advantage," Deloitte, accessed April 29th, 2020, https://www2.deloitte.com/insights/us/en/focus/cognitive-technologies/ai-investment-by-country.html

[37] Richard Stirling and Hannah Miller, "Economic Disruption and Runaway AI: What can governments do?" Oxford Insights, accessed April 29th, 2020, https://www.oxfordinsights.com/insights/worldgovernmentsummit

[38] Nagla Rizk, "Artificial Intelligence and Inequality in the Middle East: The Political Economy of Inclusion," In Oxford Handbook of Ethics of Artificial Intelligence, M Dubber, ed. (Oxford: Oxford University Press, 2020).

[39] Ellis Talton and Remington Tonar, "Unlocking the power of artificial intelligence should be a priority for infrastructure leaders," Forbes, accessed April 29th, 2020, https://www.forbes.com/sites/ellistalton/2018/07/16/unlocking-the-power-of-artificial-intelligence-should-be-a-priority-for-infrastructure-leaders/

[40] James Manyika and Jacques Bughin, "The promise and challenge of the age of artificial intelligence." Mckinsey & Company, accessed April 29th, 2020, https://www.mckinsey.com/featured-insights/artificial-intelligence/the-promise-and-challenge-of-the-age-of-artificial-intelligence

ramifications of AI in the United Arab Emirates and Saudi Arabia which recommends a policy of "upskilling": the introduction of policies to develop a highly skilled workforce for the development of AI. Upskilling here does not merely refer to the production of more data scientists but to ensure that different disciplines are engaged to "supply the needs of an increasingly voracious technology sector". It is emphasized that this can only be achieved through investment in the education sector: "investment in curriculums, in teaching staff and in the infrastructure—laboratories and research centers—needed to pursue successful AI research."[41]

Overall, the findings of this literature review enable us to observe how the identification of capacities necessary for AI development in the literature, or simply in public discourse and academic and professional conversation seems to revolve around several key areas. The undeniable importance of data access is a point emphasized in most of the works identified: AI is only as good as the data it is given. That is, the main objective of AI processes is to teach machines from experience, to 'make them smarter'. This is where the importance of data comes in: feeding AI the necessary information (data) is crucial because it affects its learning process. This area also reflects the idea that there is a specific, necessary infrastructure required for AI development. Also featured often is the role of legislation and policymaking in ensuring that AI is harnessed for the wellbeing of citizens – the emphasis on policies covers several areas, such as protecting the workforce from predicted economic turbulence accompanying AI technology, ensuring that AI technology is ethical and non-discriminatory, focusing AI capabilities towards welfare services such as healthcare and education to name a few.

IV. Research Design and Questions

The methodology for this research consists of desk research coupled with a series of semi-structured expert interviews with stakeholders in the AI ecosystem, from both the public and private sectors, across select countries in MENA. The rationale for country selection is guided by Rizk's classification of the region in terms of a spectrum of AI readiness and adoption, based on varied natural and human resource capabilities.[42] Rizk identifies leaders in the field of AI in the region as the resource-rich countries – in this study, we include the UAE as a representative of this category (characterized by large, mostly top-down investments in the field).[43]

Middle income countries in the region tend to be resource poor but are rich in skilled and educated workers. These include Egypt, Tunisia, Jordan and Lebanon. These countries have made considerable efforts in the area of new technologies such as AI and especially in terms of startups and local initiatives.[44] Yet, they are hindered by limited resources and so lag

[41] Elana Varon, "Scaling Up: The Potential Economic Impact of Artificial Intelligence in the UAE and Saudi Arabia," The Economist Intelligence Unit, May 2019, https://pages.eiu.com/May-19-Google-Public-Policy-MKT_Google-Report-Public-Policy-LP.html

[42] Nagla Rizk, "Artificial Intelligence and Inequality in the Middle East: The Political Economy of Inclusion," In Oxford Handbook of Ethics of Artificial Intelligence, M Dubber, ed. (Oxford: Oxford University Press, 2020).

[43] Nagla Rizk, "Artificial Intelligence and Inequality in the Middle East: The Political Economy of Inclusion," In Oxford Handbook of Ethics of Artificial Intelligence, M Dubber, ed. (Oxford: Oxford University Press, 2020).

[44] Nagla Rizk, "Artificial Intelligence and Inequality in the Middle East: The Political Economy of Inclusion," In Oxford Handbook of Ethics of Artificial Intelligence, M Dubber, ed. (Oxford: Oxford University Press, 2020).

behind regional leaders. We have included three interviews from Egypt, four from Tunisia as well as an interview each from Jordan, the UAE, and Lebanon. We have also secured an interview from Palestine to give the perspective of a low income, resource poor country in MENA stifled by political turmoil. A total of 11 interviews were conducted by our research team.[45]

Furthermore, desk research was conducted to give an overview of different existing attempts at mapping artificial intelligence activities in the MENA region. This includes indices as well as other methodologies including scaping of the landscape in terms of players or according to certain criteria. Given the nascent state of research, especially empirical, grounds up research on the subject in MENA, semi-structured, exploratory expert interviews were chosen to capture primary data with the objective of better understanding the state of AI in the selected countries. Questions were developed by A2K4D to gain insights into issues highlighted through the center's on-going activities and research on the subject in addition to relevant literature. To capture a sample of perspectives from stakeholders, the aim was to interview, when possible, representatives from academia, government, civil society and startups.

Guided by an ongoing objective across A2K4D research to shed light on efforts and realities of smaller players and to study the entrepreneurial and grounds up efforts in MENA, where possible startups, were selected to represent the business community in this study. Interviewees were asked about the state of AI in their respective countries, about the difference in such state between the public and the private sectors, about the importance of data to the state of AI, about data openness, and about their knowledge of AI for development. Additionally, opinions were sought to assess the necessary capacities to make the most of new technologies, the role governments should play in supporting AI and AI for development, and the role other actors can play in the AI ecosystem. Whenever possible, the questions were designed to examine local perspectives on related challenges, as well as gain local insight as to the way in which these challenges ought to be tackled.

While we set off with the objective of securing an equal number of interviews from each of the countries included in the study, we were unable to do so. This was an anticipated challenge, given the unprecedented circumstances of 2020 and the global COVID-19 pandemic. Nevertheless, to capitalize on the wealth of qualitative information and insights from the interviews, the research team devised a short survey, with the aim of soliciting a wider cross section of quantitative responses across the region. This is carried out in an attempt to engage with a larger sample and collect quantitative data to complement the in-depth insights from the interviews. Results obtained from the survey will later be used to produce infographs to complement and follow up on this research.

Table 1: Information about interviewees

Interviewee Name	**Position**	**Country of Affiliation**

[45] Nagla Rizk, "Artificial Intelligence and Inequality in the Middle East: The Political Economy of Inclusion," In Oxford Handbook of Ethics of Artificial Intelligence, M Dubber, ed. (Oxford: Oxford University Press, 2020). For information regarding interviewees, please see the table at the end of this section.

Jazem Halioui	Founder & CEO of WebRadar	Tunisia
Hatem Haddad	Co-Founder and CTO of iCompass	Tunisia
Lilia Ennouri	Co-founder of Data Co-Lab	Tunisia
Alaa Triki	Co-founder of Data Co-Lab	Tunisia
Issa Mahasneh	Executive Director of the Jordan Open Source Association	Jordan
Abed Khooli	Data scientist and researcher	Palestine
Ahmed Abaza	Founder and CEO of Synpase Analytics	Egypt
Ayman Ismail	Associate Professor at the American University in Cairo (AUC) and the founding director of AUC Venture Lab, AUC Angels and AUC Innovation Hub.	Egypt
Golestan Radwan	AI Advisor to the Minister of Communications and Information Technology	Egypt
Nikolaos Mavridis	Founder and director of the Interactive Robots and Media Lab	United Arab Emirates
Bahia Halawi	Co-Founder of Data Aurora	Lebanon

V. Findings and Discussion:

This section reflects on the key findings from desk research and interviews undertaken for the paper. Questions will inquire as to changes interviewees would like to see with regards to data usage and production and their opinions on the pertinence of more open forms of data. The interviews will also probe views on necessary capacities for AI for development to flourish in MENA. The role of government and policymakers will also be explored, as well as the role of other actors in the ecosystem.

A. What are existing narratives and mappings of AI in the MENA region?

As AI remains relatively novel in MENA, few attempts exist at mapping relevant issues and the status of AI in the region. For the purpose of this research, several of those attempts were surveyed. Given the nascent nature of research on the topic regionally and the many questions that persist regarding the expected trajectory of AI in MENA, it became apparent that there is a need for further empirical research and mapping. As such, A2K4D surveyed existing mappings that we are aware of in attempt to identify further aspects to document and map. In the following, we list these attempts, describing the kinds of methodologies, if any, they employed.

Table 2: Survey of Existing AI Mappings Covering MENA Countries

Report name	**Year**	**Countries covered**	**Included in this study**	**Publisher**

'Meet the 100 Arab start-ups shaping the Fourth Industrial Revolution'	2019	Arab states (minus Sudan, Somalia, Mauritania)	Egypt, Jordan, Lebanon, UAE, Tunisia, Palestine	World Economic Forum
'Government AI Readiness Index'	2019	Global	Egypt, Jordan, Lebanon, UAE, Tunisia	Oxford Insights
'Middle East prepares for AI acceleration: Exploring AI commitment, ambitions and strategies'	2019	UAE, Egypt, Saudi Arabia and Qatar	UAE, Egypt	IBM
iHub Review of Egypt-based Start-ups	2020	Egypt	Egypt	Ain Shams
'Mapping the Data Economy Landscape in the Middle East and North Africa (MENA): Key Stakeholders, Opportunities, Challenges and Recommendations'	2020	Arab states	Egypt, Tunisia, UAE, Jordan, Palestine	Birzeit University Center for Continuing Education
'Global South Map of Emerging Areas in AI by Knowledge for All'	2018	Global South (Asia, LAC, MENA, and SSA)	Egypt, Jordan, Tunisia	Knowledge4All

- *Meet the 100 Arab start-ups shaping the Fourth Industrial Revolution by WEF*[46]

Since this is a 'listicle', methodology does not play much of a role – but it is stated in the report that the start-ups listed were selected from around 400 applicants. The startups belong to a number of sectors, including education, energy, environment, finance, health and the media.

[46] Abdulrahman Tarabzouni, Ahmed El Alfi, Amir Farha, Areije Al Shakar, Hala Fadel, Khaled Talhouni and Mirek Dusek, "Meet the 100 Arab start-ups shaping the fourth industrial revolution", World Economic Forum, accessed November 25th, 2020. https://widgets.weforum.org/arabstartups/

- *2019 Government AI Readiness Index*[47]

The Government AI Readiness Index, produced with the support of the International Development Research Centre (IDRC), is the second edition of the index, which aims to measure to what extent governments around the world are ready to use AI in the delivery of public services. The overall score of each country is comprised of 11 input metrics, grouped under four high-level clusters: governance; infrastructure and data; skills and education; and government and public services. The data is derived from a variety of resources, ranging from desk research into AI strategies, to databases such as the number of registered AI startups on Crunchbase, to indices such as the UN eGovernment Development Index. Although the 2019 edition sees an expansion of scope to cover all UN countries, it does not pay particular attention to the Middle East region with regard to its methodology.

- *Middle East prepares for AI acceleration: Exploring AI commitment, ambitions and strategies*[48]

This report explores the ways in which AI has progressed in several of what it identifies as "forward-looking countries in the Middle East". To gain insight into the field, this report was produced through a methodology of surveying 200 executives from across the four countries, as part of a broader survey of 5,000 executives globally. According to the report, responding executives included "C-suite members"—CEOs, CMOs, CFOs, COOs, CIOs and CHROs—as well as heads of customer service, information security, procurement, product development, sales functions, and others.

This potentially reflects an interesting contrast to our approach at A2K4D, which is more bottom-up and pays greater attention to smaller players, not only domestically (focusing on the start-up scene in Egypt rather than corporations, for example), but also geopolitically as well (looking at often overlooked countries with respect to AI development in MENA, such as Tunisia for example).

- *iHub Review of Egypt-based Start-ups*

This review is a regularly updated infograph of sorts created by Ain Shams University affiliated innovation hub, iHub. The infograph contains a mapping of several start-ups which employ AI technologies in their operations. Start-ups were divided into several categories, such as E-commerce, FinTech, Ed-Tech, Agri-Tech, Health-Tech, Food and Grocery, Delivery and Transport, Travel, and Energy. The methodology of the research is clearly stated, but it appears to be crowd-sourced, with iHub's website providing a page through which submissions to include other start-ups could be made to fill in what was missed by the infograph. This could be an example of a bottom-up approach of mapping, although it leaves room for interpretation with regard to what constitutes the definition of a start-up.[49]

[47] Hannah Miller and Richard Stirling, Government Artificial Intelligence ReadinessIndex 2019, Oxford Insights, accessed 14th October 2020. https://www.oxfordinsights.com/ai-readiness2019

[48] Ian Fletcher, Brian Goehring, Anthony Marshall, and Tarek Saeed, "Middle East Prepares for AI Acceleration: Exploring AI Commitment Ambitions and Strategies," IBM Institute for Business Value, November 2019, https://www.ibm.com/downloads/cas/XWPEAP0V

[49] Innovation Hub, "Egyptian Tech Startups 15/8/2020", iHub, May 2020. http://ihub.asu.edu.eg/egystartups.html

- *Mapping the Data Economy Landscape in the Middle East and North Africa (MENA): Key Stakeholders, Opportunities, Challenges and Recommendations*

This report is perhaps more ambitious than others listed in this section in its scope. In its education section, the report features a list of higher education programs that relate to AI. However, the report is not concerned with AI specifically, but rather with 'new and emerging technologies'. Perhaps for this reason, legal frameworks and policy strategies relating specifically to AI do not feature in the report, and its survey of start-ups does not exclusively cover AI startups. Startups are instead classified by industry (e-commerce, fashion, entertainment, etc.; not based on whether or not they employ AI).

- *Global South Map of Emerging Areas in AI by Knowledge4All*

The mapping process in this project identifies Artificial Intelligence (AI) emerging markets in four regions: Asia, LAC, MENA, and SSA. The methodology uses what the researchers call "manual mapping", a bottom-up approach which utilizes a larger researcher network, in addition to partner sites and City.AI community in each country/region. The mapping emerged based on several elements. One of these elements includes LinkedIn search results "which show the total number of profiles according to their own specialized parameters". The authors also refer to a survey of the names of leading AI conference presenters who "were considered to be influential experts in their respective fields of AI". Finally, researchers looked at other additional "reports and anecdotes from the global community". The authors of the webpage note several limitations to the mapping process. One such limitation is the subjectivity of the mapping of the country entities, since researchers rely on input from country experts and could reach out to only a few experts per country in the time allowed. Another limitation mentioned is the extent to which judgments of what constitutes AI are taken at face value, despite research efforts to verify country expert credentials.

Comments

As demonstrated above, we were unable to find evidence of a streamlined approach with regard to the methodologies used in these various mapping efforts. This speaks to a larger issue in the field with regard to methodology as well as the need for more transparent research and mapping efforts with respect to AI.

B. AI Strategies in MENA

Figure 4: AI Strategies in MENA

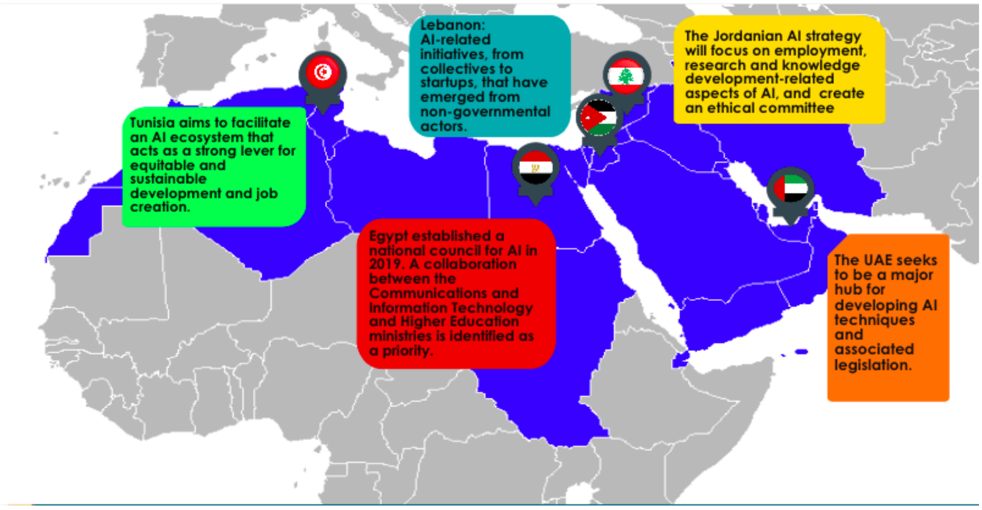

Figure created by A2K4D team based on data retrieved from:
Egypt: A2K4D collaborated with local civil society actors to propose a roadmap towards an inclusive national AI strategy for Egypt.
Tunisia: "National AI Strategy: Unlocking Tunisia's Capabilities Potential," Agence Nationale de la Promotion del la Recherche Scientifique, September 2018, http://www.jaist.ac.jp/~bao/AI/OtherAIstrategies/National%20AI%20Strategy:%20Unlocking%20Tunisia's%20capabilities%20potential.%20–%20Agence%20Nationale%20de%20la%20Promotion%20de%20la%20Recherche%20scientifique.pdf
IIG working paper, A2K4D IIG working paper, A2K4D
UAE: Anna Zacharias, "UAE Cabinet Forms Artificial Intelligence Council," National News UAE, March 2018, https://www.thenational.ae/uae/uae-cabinet-forms-artificial-intelligence-council-1.710376, Dom Galeon, "The United Arab Emirates is the first country in the world to hire a minister for artificial intelligence," Business Insider, December 2017, https://www.businessinsider.com/world-first-ai-minister-uae-2017-12
Jordan*:* JOSA, *'About JOSA',* accessed November 2020, https://jordanopensource.org/about-u
Lebanon: Interview with Bahia Halawi, October 2020.

In recent years, and in light of rapid developments in the field of AI, a number of states have taken steps toward implementing a national AI strategy outlining the countries' vision for the future and impact of AI. There are currently more than 20 countries that have released strategy documents about AI. The list includes Canada, China, France, the European Union, India, Kenya, and the United Arab Emirates.[50] In the following, we look at the main pillars of existing AI strategies for countries covered in this study.

1. Egypt

In its Vision 2030 agenda, it is indicated that Egypt aspires to implement AI in the fields of development and economic growth.[51] Towards that end, in April of 2019 the Ministry of Communications and Information Technology (MCIT) announced the appointment of Golestan Radwan as the Ministry's advisor for AI.[52] Additionally, Prime Minister Mostafa Madbouly issued a decision announcing the establishment of a national council for AI, to be

[50] A2K4D, "Why is a National AI Strategy Important for Egypt," Knowledgemaze, April 2019, https://knowledgemaze.wordpress.com/2019/04/15/why-is-a-national-ai-strategy-important-for-egypt/
[51] Ministry of Communications and Information Technology, "Artificial Intelligence: Egypt's AI Strategy", accessed November 28th 2020, https://mcit.gov.eg/en/Artificial_Intelligence
[52] Telecompaper, "Egypt's ICT ministry appoints Radwan as AI advisor", tp:news, April 2019, https://www.telecompaper.com/news/egypts-ict-ministry-appoints-radwan-as-ai-advisor--1287278

chaired by the Minister, Amr Talaat, in addition to an executive office within the council (headed by eight experienced members nominated by the council's various members).

Also in 2019, Khaled Abdel Ghaffar, Egypt's Minister of Higher Education, stressed that the country is currently prioritizing collaboration between the two ministries (Higher Education and MCIT). As such, the Ministry has collaborated with MCIT to add AI departments and to equip the youth and curriculum to meet the current needs of the market. For example, the country now has the first AI university that is specialized in AI in Kafr El Sheikh, which was established in collaboration with the UAE.[53]

As for the AI strategy itself, it is anticipated that the focus will be on promoting specific sectors, such as agriculture, medicine and language processing.[54] The strategy is also expected to focus on the reskilling the Egyptian workforce to adapt to future technological changes. On his part, Talaat has stressed Egypt needs a legal framework to address these changes, and since data is the focal point of AI, that the country needs the legal frameworks to focus on issues of data privacy and protection.[55] However, at the time of writing, the complete strategy remains confidential, despite input volunteered by several private stakeholders.[56]

2. *Tunisia*

Like its Egyptian counterpart, the Tunisian government has yet to reveal the release date for the strategy. Due to the early stages of Tunisia's Strategy it was impossible to provide information on format, ethics, infrastructure, community engagement, or AI definition.[57] Available information however provides that Tunisia's national strategy for AI is titled 'National AI Strategy: Unlocking Tunisia's Capabilities Potential', and was developed following a workshop hosted by the UNESCO Chair on Science, Technology, and Innovation Policy, in partnership with the National Agency for Scientific Research Promotion-ANPR, in 2018.[58]

3. *UAE*

Titled *Artificial Intelligence Strategy 2031,* the UAE strategy for AI was initiated in October, 2017 as a major initiative within the UAE Centennial 2070 objectives.[59] The UAE also

[53] Egypt Today staff, "First ever Egyptian AI faculty inaugurated", Egypt Today, July 2019, https://www.egypttoday.com/Article/1/73007/First-ever-Egyptian-AI-faculty-inaugurated
[54] Ministry of Communications and Information Technology, "Artificial Intelligence: Egypt's AI Strategy", accessed November 28th 2020, https://mcit.gov.eg/en/Artificial_Intelligence
[55] Ministry of Communications and Information Technology, "Artificial Intelligence: Egypt's AI Strategy", accessed November 28th 2020, https://mcit.gov.eg/en/Artificial_Intelligence
[56] A2K4D was involved in contributing to Egypt's national AI strategy and has submitted a draft proposal. No response or update was given as to the status of the strategy to date.
[57] Jelel Ezzine, "National Al Strategy: Unlocking Tunisia's capabilities potential", Agence Nationale de la Promotion de la Recherche scientifique, May 2018, https://www.slideshare.net/Jelel.Ezzine/national-ai-strategy-unlocking-tunisias-capabilities-potential.
[58] "National AI Strategy: Unlocking Tunisia's Capabilities Potential," Agence Nationale de la Promotion del la Recherche Scientifique, September 2018,
[59] IIG working paper, A2K4D

appointed its first Minister of AI in 2017, becoming the first country to do so.[60] A council for AI was also formed in 2018 to implement the technology in areas alongside the public sector and public transport (two of the first sectors to implement AI in the country).[61] However, very little public information is available with regard to the details of the strategy and its methodology.[62]

Available information reveals that the strategy was designed to mark the UAE as a hub for developing AI techniques and associated legislation, in order to generate new revenues and provide fresh opportunities for the national economy.[63] While Tunisia's strategy is expected to tackle job scarcity, the UAE's appears to be more wide-ranging, covering areas such as transport, health, and space.

4. Jordan

The Jordanian Artificial Intelligence Policy is expected to come into force toward the end of 2020.[64] The strategy will focus on employment, research and knowledge development-related aspects of AI, as well as creating an ethical committee for AI to discuss ethical implications of the technology.[65] This prioritization of ethics is influenced by the Jordanian government's partnership with the Jordan Open Source Association (JOSA), a non-profit organisation based in Amman, registered under the country's Ministry of Digital Economy and Entrepreneurship.[66]

5. *Palestine*

Palestine does not have a national AI strategy to date. As highlighted by our interviewee Abed Khooli, Palestine, as a country 'under occupation with no control over resources, economy or borders', enjoys a public sector approach that he describes as 'verbal welcoming' (that is, encouragement limited to statements and PR activities). However, on the policy level, no strategy for AI has been drafted or implemented at the time of writing.

6. *Lebanon*

Lebanon does not have a national AI strategy to date. AI initiatives exist, as evidenced by our interview responses, but there are no governmental initiatives. Yet there are a number of AI-related initiatives, from collectives to startups, that have emerged in the country from

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[60] Dom Galeon, "The United Arab Emirates is the first country in the world to hire a minister for artificial intelligence," Business Insider, December 2017, https://www.businessinsider.com/world-first-ai-minister-uae-2017-12

[61] Anna Zacharias, "UAE Cabinet Forms Artificial Intelligence Council," National News UAE, March 2018, https://www.thenational.ae/uae/uae-cabinet-forms-artificial-intelligence-council-1.710376

[62] IIG working paper, A2K4D

[63] The National, "UAE looks to artificial intelligence strategy to prepare for the future". The National News UAE, May 2018, accessed November 26th 2020, https://www.thenationalnews.com/uae/uae-looks-to-artificial-intelligence-strategy-to-prepare-for-the-future-1.667749

[64] Interview with Issa Mahasneh

[65] Ibid.

[66] JOSA, 'About JOSA', accessed November 2020, https://jordanopensource.org/about-us

non-governmental actors. As our interviewee Bahia Halawi pointed out, however, such initiatives ‘do not match up to what would be a governmental effort’.[67]

VI. Interview Results

The section below consists of synthesized results from the following interviews : Three interviews conducted in Egypt with Ahmed Abaza, CEO of Synapse Analytics, Golestan Radwan, Advisor to the Minister of Communications and Information Technology, and Ayman Ismail, founder and director of the AUC Venture Lab. Regional interviews with Hatem Haddad, Co-Founder and CTO at iCompass in Tunisia, Alaa Triki and Lilia Ennouri (from Tunisia-based Data Co- Lab), Jazem Halioui (founder and CEO of WebRadar in Tunisia), Nikolaos Mavridis, founder and director of the UAE-based Interactive Robots and Media Lab, Abed Khooli (Birzeit University in Palestine, Data Science, Data Journalism and OpenData Specialist), Issa Mahasneh (Executive director, Jordan Open Source Association) and Bahia Halawi (Co-Founder of Lebanon-based Data Aurora).

A. What is the state of AI in MENA? Are we producers or adaptors? What examples can you provide of AI applications.

Like other countries in the region, the interviewees for Egypt affirmed that Egypt is an adopter of AI technologies existing elsewhere. As noted by Golestan Radwan, Egypt is directing efforts towards identifying priority areas and customizing solutions, being one of the first countries in the MENA region to introduce a national AI strategy. Through this strategy, the country hopes to tailor solutions to local developmental needs rather than adopting solutions ill-suited to deal with Egypt’s priorities. Radwan also shared the view that the AI scene in the MENA region is very diverse. There are countries that identified AI as a national priority, while others have yet to tackle the issue. She added that countries in the MENA region also have different levels of awareness, readiness of available data, infrastructure and tech literacy. Ayman Ismail pointed to the limited deployment of AI in Egypt due to the underutilization and lack of access to datasets. In Egypt specifically, Radwan explained that official efforts are being directed at identifying priority areas and customizing solutions to local development needs, highlighting how using ‘off-the-shelf products’ is problematic.

Popular applications of AI in Egypt, according to Ahmed Abaza, involve machine learning algorithms, which are used in fields such as e-commerce platforms and credit scoring. In Egypt and the MENA region, algorithms are used to develop and manufacture recommendations on platforms such as Souq. Abaza added that adapting machine learning for alternative means of credit scoring is the most popular use of the technology, since 90 percent of the population in Egypt is not integrated into the formal banking system. He added that Egypt lacks quantum computing and 5G networks, which are two necessary enabling technologies for AI to flourish at a wider scale. On the regional level, Abaza noted that a popular use of AI can be found in smart cities: autonomous systems, fire detection, and traffic management, to name a few examples. Finally, Abaza noted that Natural Language Processing in Arabic is a big race in the MENA, with several universities in the region currently working on it.

[67] Interview with Bahia Halawi, October 2020

The four Tunisian interviewees had consensus that Tunisia is both an adopter and producer of AI, however with production being in the preliminary stages. Jazem Halioui stated that AI is especially attractive to engineers, many of which are self-taught via the Internet. He added that machine learning is the most prevalent AI technology in Tunisia. Hazem Haddad highlighted that AI technologies adopted are tweaked and tailored to the Tunisian context, adding deep learning to machine learning as the most prevalent types of AI in Tunisia. Lilia Ennouri explained that there is a rise in AI-based products in Tunisia, led by the private sector. Alaa Triki noted the difference in the level of AI implementation among MENA countries, with the United Arab Emirates (UAE) leading in terms of governmental AI initiatives. The Tunisian Startup Act, implemented in 2018, provided benefits and incentives to entrepreneurs to encourage them to innovate in the use of AI.

Issa Mahasnah from Jordan stated that the country has a good startup scene across different fields, among them AI, the data economy, and data governance. He added that most startups in Jordan focus on neuro-linguistic programming (NLP), giving the example of Mawdou3, an online Arabi content publisher.[68] Other examples include Oujeeb, a digital platform for Q&As and Labiba, an Arabic chatbot building company.[69]

As for Palestine, Abed Khooli stressed that AI is getting more traction in light of COVID-19 due to the increased need for digital transformation, with AI adoption being the main component. Machine learning is also in wide use in the region, examples being RedCrow, Mashvisor, YaMsafer, Imagry, WebTeb/Wekaya.[70] There is also some chatbot development in Palestine.

Nikolaos Mavridis started the first robotics lab in the wider region of the Gulf in 2007, Media Lab in the UAE. He shared the view that the MENA region countries are both consumers and adaptors of AI, not producers. The region produces data, but does not produce the codes or algorithms for AI. And while there is innovation in applying AI, it is not innovation in creating AI. The UAE has numerous government-funded initiatives and competitions centering around AI. Two examples are the drones for good competition and the robotics and AI for good competition.[71]

Bahia Halawi stated that Lebanon has some non-governmental AI initiatives, but does not have an AI strategy. The AI initiatives in Lebanon are related to media and advertising, targeting markers in the UAE, Saudi Arabia, and Kuwait. For the younger generation, there are AI bootcamps where students are introduced to AI implementations, such as the Internet of Things (IOT). Halawi mentioned the exception of a few academic papers with algorithmic theoretical knowledge focusing on AI existing.

[68] See https://mawdoo3.com
[69] See https://www.labiba.ai/about-labiba
[70] See https://www.redcrow.co/; https://www.mashvisor.com/; https://www.webteb.com/
[71] Steve Flynn, “The UAE Drones for Good Award 2017 Winners and Finalists”, Skytango, April 2017, https://skytango.com/the-uae-drones-for-good-award-winners-finalists/;The National Staff, “UAE AI and Robotic Award for Good winner to receive $1m”, The National, June 2015, https://www.thenationalnews.com/business/technology/uae-ai-and-robotic-award-for-good-winner-to-receive-1m-1.96219

B. Does the state of AI and new technologies differ in the public vs. private sector? How?

Among the MENA countries examined, interview respondents all concurred that the state of AI and new technologies differs in the public vs. the private sectors. Radwan explained that globally the adoption and production of AI has been driven by the private sector. Companies such as Tesla, Amazon and Google have historically led AI development, and this continues to be the case in developed countries. For example, the Unites States does not have a national AI strategy. The applications of AI in such a setting (for example, product recommendation) are not attractive the governments, and are highly economically valuable to private actors. The MENA region presents a different scenario, with attractive AI uses in the public sector. Egypt is aiming to become a pioneer in showcasing AI-drive use cases within the public sector.

Abaza pointed to the fact that the public sector is always slower than the private sector, yet the public sector needs to avail resources for the private sector to adopt. It is his view that AI in the private sector will be adopted faster, since private companies are investing more in AI and have better infrastructure and data management tools. In Egypt, the delay in standardization is an advantage, since it allows for adaptation to new AI. Abaza noted that the Ministry of Communications and Information Technology (MCIT) is playing an active role in promoting digitization and AI, including offering learning courses. Abaza noted that his AI services company faces the challenge of the reluctance of other companies to work with them. Ismail noted that, for the private sector, the focus is mainly in telecommunications companies and banks. As for the public sector, he said he was only aware of the initiative of establishing a national AI strategy for Egypt.

For Tunisia, Haddad described the state of AI in the public and private sectors as “two speed roads”: one is the highway (the private sector) and the other is a slow one with a lot of barriers (the public sector). Ennouri reiterated the view that governmental use of AI is non-existent, and that the government does not save data to use in AI. Data Co-Lab is lobbying with the government to work in that area, in addition to offering training sessions on data, machine learning, and AI. In the private sector, startups and companies request data consulting in order to know how to manage data better and enhance customer experience. Triki shared the similar view that in Tunisia there is a huge gap between usage of AI in the public and private sectors. The private sector relies on specialized engineers, while the public sector is mainly focused on collecting data and encouraging actors to share their data via the open data portal initiative.[72] Halioui stated that in the private sector many startups claim to use AI and data analysis in different configurations and environments, such as language processing and analyzing large quantities of textual data. Other uses include AI in the financial sector for credit scoring and in marketing to tabulate and influence consumer behavior.

In Jordan, Mahasneh explained that the private sector is ahead in terms of implementing AI initiatives and using new technologies, yet the public sector has progressive policies and strategies, including laws related to new technologies. Resource availability is one of the

[72] The Republic of Tunisia, Open Data Portal, accessed November 2020, http://www.data.gov.tn/

reasons for the gap in implementation between the private and public sectors, in addition to skillset issues and more attractive pay in the private sector. Nevertheless, there have been AI initiatives implemented by the public sector. One example executed by the Ministry for Social Development is the National Unified Registry (NUR), which merges different datasets from different entities within a single database and carries out data analysis based on that database.[73] Furthermore, the Jordanian government has plans to implement a national information system, although this has not materialized beyond written plans to date. Mahasneh pointed to several obstacles facing AI in Jordan, such as difficulty in obtaining public sector data sets and data ownership issues. There are also international companies based in Jordan which use AI for logistics, such as Aramex and Souq.

For Palestine, Khooli explained that the private sector, and especially small enterprises (which comprise over 9 percent of economic activity), is ahead of the public sector. A number of universities started to offer data science courses, and a few startups have AI (machine learning or data analytics) as part of their business model. Consumer behavior is the main area in which AI is being deployed.

Mavridis shared the view that for the UAE, technologies related to explainable AI are more important to the public sector. He stressed the dire need for AI to augment, rather than replace, humans. AI in the public sector has various applications, such as calculating risk factors, security and biometrics in airports, and detection and prediction for hospitals (time slots). AI uses also extend to efficiency and performance, consumer satisfaction and cost, health, and insurance systems.

In Lebanon, Halawi stated that the private sector is always ahead with interdisciplinary work, while the public sector is risk averse and awaits to monitor if AI initiatives are beneficial to the private sector and then follow suit.

C. How important is data to the state of AI in your country/ MENA? What issues do you face in terms of data use or production?

Interviewees were asked about the importance of data, and about obstacles and challenges they face or are aware of in relation to data and AI.

For Egypt, Radwan mentioned that the quality and relevance of data are key to the state of AI. The ability to identify the right type of data to realize applications of priority is vital, and so is the type of infrastructure necessary to support that. Abaza stressed the importance of having open data sets available, such as consumer or banking data (which can be used to for credit scoring algorithms). He also shared the view that the government should play an active role in making data available for the benefit of the ecosystem. In Abaza's viewpoint, China's success with regards to AI is largely due to the access to data the country has. Ismail mentioned that in Egypt startups rely on their own data, citing the examples of Paymob, Vezeeta, Souq, and Jumia. However, there are no new products being developed as a result of this data accumulated, and it is unclear how the data is utilized by startups. Egypt is in the primitive stages of product development, despite the prevalence of big data analytics for

[73] Hashemite Kingdom of Jordan, "National Social Protection Strategy 2019-2025", UNICEF, https://www.unicef.org/jordan/media/2676/file/NSPS.pdf

existing products and targeted advertising. Ismail stipulated that the challenge with data is that corporates want to restrict access to their data due to privacy concerns, legal issues, and to maintain a competitive edge. Echoing Abaza's frustration at the lack of available local data sets, Ismail suggested a solution, whereby university research labs work on large anonymized datasets provided for research purposes. The government is also protective of its data in Egypt, according to Ayman.

From the Tunisian perspective, Halioui stressed that data is fundamental for AI and machine learning, since machine learning consumes data. Ennouri affirmed this notion, stating that there is no AI if there is no data, especially when it comes to machine learning and deep learning. It would be increasingly difficult to work on AI products if the needed data is not available. Triki also shared this view, adding that quality data in quantity is needed for AI. Haddad stated that from a personal point of view, since he works in deep learning, there is a dire need for data. Additionally, private and governmental institutions are very skeptic about giving their data to AI companies, which poses a big challenge.

Like many other MENA countries, data collection in Tunisia faces several obstacles. Triki explained that corporation do not share their data, and if data is shared it is in an unusable format. Furthermore, infrastructure barriers in relation to cloud storage services for computing and collecting big data remains an issue. And although legislation surrounding data openness exists in Tunisia, implementation of such legislation is poor. For example, Marsoum 54 of 2011 makes it mandatory for any public administration to publish public data online.[74] However, as reiterated by both Halioui and Haddad no one knows how to apply the legislation surrounding data. Additionally, there are no rules on how to collect, govern, and use private data. Ennouri also mentioned the problem of data poverty in MENA. Halioui added that the culture of open data is still young in the MENA region when compared to other regions of the world, partially due to social and political contexts. However, since 2011 the region has been experiencing more openness of society and politics, a phenomenon which seems to correlate with open data. In Tunisia, public data is not open and accessible to everyone.

If the importance of data to the state of AI were to be ranked from 1 to 10, Mahasneh would score it as 8. Nevertheless, there is little data sharing in Jordan. A 1972 law continues to govern data in Jordan, which stipulates that anything not public or classified is illegal to release.[75]

Khooli from Palestine shared the view that appropriate datasets are the core of AI, and thus if the importance of data to AI is to be ranked, he would rank it as 10 if not even more. Both the public and private sector can utilize the same datasets, albeit for different purposes. Khooli shared issues related to data in Palestine, which include data availability, access to data (discovery, technical and legal), data quality (accuracy, completion, updates), cost (if data is

[74] See Khouloud Dawahi, "Tunisia breaks down government's secrecy walls", freedominfo.org, June 2016, http://www.freedominfo.org/2016/06/tunisia-breaks-down-governments-secrecy-walls/
[75] See Jordan Open Source Association (JOSA) and Privacy International, "The Right to Privacy in the Hashemite Kingdom of Jordan", Stakeholder Report Universal Periodic Review 31st Session – Jordan, March 2018, https://uprdoc.ohchr.org/uprweb/downloadfile.aspx?filename=5856&file=EnglishTranslation

to be collected or acquired), and infrastructure (storage, compute and networks). Capacity is also required, and not enough data science and data journalism expertise are available.

Mavridis explained that a wide variety of technology, including AI, depends on data. And like Radwan, he added that although large amounts of data are needed, the kind(s) of data is vital. The importance of data stems from the use of this data. And although it is preferred to use anonymized data, Mavridis stated that AI can potentially reveal identities, which leads to privacy concerns. As for the obstacles and issues surrounding data, Mavridis raised the points of how much data will ever be considered enough, and should data be regarded as a personal asset or as property.

According to Halawi from Lebanon, without data, AI is not applicable, it remains theoretical. In order to have promising insights from AI, local, contextualized and updated data sets are needed. Halawi said that among the issues with data is that there is no open data portal in Lebanon, which prevents transparency. Unfortunately, in Lebanon there is no heightened awareness of the importance and potential impact of data.

D. *What are the region's top three most pressing developmental concerns? And could AI play a role in mitigating these issues? And what does AI for development mean to you?*

In Radwan's opinion, AI for development is a tool to solve developmental problems, and an opportunity to get youth into the global labor market. AI presents leapfrog potential for companies, and countries, in terms of innovation in key sectors such as agriculture, infrastructure, water management, healthcare, economic planning, and education. In Abaza's viewpoint, invisibility in Egypt's economy is a big problem for development, and AI can play a role in visibility of the money in the country and understand people's behavior in novel ways. For example, machine learning can tell us estimates of people's wealth including their informal spending patterns and money flow. This kind of visibility has the potential to attract investors and provide opportunities for the country. Health is another priority area, and health data is very important. In Egypt, the nation-wide campaign to eradicate virus C has collected large amounts of health data, which could be used for preventative medicine. According to Ismail, Egypt's top priority developmental issues are poverty, equity, jobs, and water access. And unless accompanied by protective policies, technologies amplify existing inequalities in societies.

Halioui shared the viewpoint that if AI focuses on development, then it is AI for development. The most pressing developmental concern for Tunisia, in his viewpoint, is human wellbeing, specifically depression. In Tunisia, there is an application called Fabulous that works on self-development, and helping individuals self-heal, change their habits, and develop self-awareness.[76] Haddad stated that for Tunisia the priorities in terms of development are employment, education, and health, adding that startups can help solve all these issues. In fact, startups in AI can offer various employment opportunities for youth in Tunisia. According to Ennouri, there are three pressing developmental concerns in Tunisia:

[76] Emna Ghariani, "Feeling Fabulous: Tunisian app helps healthy 'rituals', Tought Leadership: Wamda's features, Opinions, and Podcasts, November 2016, https://www.wamda.com/tag/Fabulous

education, health (especially in light of COVID-19), and transportation. For example, in health, AI can play a role in expenditure saving by detecting diseases.

Triki explained that new technologies, like big data and AI, can help with data collection and analysis, which can then feed algorithms that help to more accurately design programs to mitigate developmental issues. He listed three main issues for Tunisia: unemployment, poverty, and freedom of expression. For unemployment, data of the level of education, expectations, expectations, and preferred field of work need to be collected, and later developed into a project that meets the needs of the unemployed. For poverty, data such as income and location could be collected and analyzed, to help determine if there are poverty hotspots and create local strategies to combat them. Unfortunately, AI will not be able to mitigate issues such as freedom of expression, which requires legislative change.

In Jordan, Mahasneh explained that the government drafted an AI strategy, which should be approved in the near time frame, which focused on academic aspects but also took into consideration the ethical concerns of AI.[77] As recommended by Jordan's Open Source Association (JOSA), of which Mahasneh is the executive director, the strategy includes creating an ethical committee for AI to discuss the ethical implications of the technology. Unemployment is one developmental concern for Jordan mention by Mahasneh, and there is a need to match the capacities of higher education graduates to the demands of the labor market. There is a fear that in the public sector, AI could replace certain human-centric roles, as was the case with the introduction of e-governance.

AI for development means using data enabled products, services or processes to add economic or social value, according to Khooli from Palestine. The three developmental concerns identified by Khooli are health, education and economy, with cross cutting issues like freedom and equality. In his view, new technologies are double-sided tools, they may help or they make things worse. The three previous industrial revolutions did not end disparities in societies. And while it is true that current technologies are improving access to marginalized groups, or people who are restricted by social norms, this is not enough. Khooli explained that there is a risk that products or services not observing data ethics may duplicate bias and widen the digital gap. To overcome this, regulations, legislation, policies, and capacity building for AI workers is needed.

Mavridis explained that AI for development involves special programs which help forward existing developmental roles. He clarified that data for development, for example supervised machine learning, on its own, is not a solution, but a prerequisite for AI for development. The United Nations Millennium Development Goals (MDGs) and Sustainable Development Goals (SDGs) can be directly connected to AI, in fact, the MDGs support the link between AI and development. The Poverty Action Lab hosted at MIT shows the importance of evidence-informed policies and real-time systems that use data science and are AI-enhanced

[77] Prime Minister Omar Razzaz has recently approved the formation of a national committee on artificial intelligence (AI) to oversee the introduction of the technology into different governmental economic sectors. See Maram Kayed, "Gov't to harness artificial intelligence for boosting services – ICT minister", The Jordan Times, Nov 2019, https://www.jordantimes.com/news/local/govt-harness-artificial-intelligence-boosting-services-%E2%80%94-ict-minister

or powered. And when it comes to food shortages, data science and AI can play an important role in improving food resource allocation. As for health, AI can play a role in the early detection and disease prevention for people with special needs. AI can also play a role in waste management for cities and slums, provided the necessary data exists. When it comes to gender inequality, although there is bias in AI, there are desirable biases that can lead to affirmative action. One example is the for specific racial groups who have been historically denied access to education being given opportunities until an equilibrium is reached.

For Lebanon, Halawi stated that the top three developmental concerns are issue of refugees, corruption, and socioeconomic inequalities. Socioeconomic inequalities include gender issues and gaps between rural and urban areas.

E. Data openness: what does it mean? What role can it play for AI and development? Concerns with data openness?

Abaza emphasized the importance of data openness. He explained that one private company in Egypt can have access to 70 percent of medical data on Egyptians, causing a data monopoly. There is a dire need for anonymized data, added Abaza. He gave the example of the New York open data set[78], which can be used by potential restraint owners to check traffic figures, etc. Abaza states that he feels "stuck" when it comes to consent and anonymity with data, because although privacy concerns are legitimate, the more data available, the more his company can avail products to the market. Ismail added that we need to build local datasets in order to solve local problems, in order to contextualize AI rather than apply exported AI. Radwan cautioned that sweeping statements about data openness are problematic. She proposed as an alternative approach: data governance policies and strategies, both of which are tackled in Egypt's AI strategy, and are an important focal point rather than simple openness. Radwan reiterated that the key lies in identifying real world problems and necessary applications. Based on this, countries can develop strategies which involve technology, infrastructure and policies, which will ensure data is in the right and required form and respects the needs and rights of people.

Halioui from Tunisia explained data openness as "a form of trust between people." Data is essential for AI: the more data available the more applications can be created. However, Halioui added that along with the notion of data openness, one must be aware of the concept of data protection. The types of data that should be open and the types of data that should remain private is a matter to be decided on a case-by-case basis. Haddad added that data is needed to feed deep learning models, and this includes access to quality open data. And because his company does big data analytics on social media data, Haddad often faces the problem of privacy. There are no clear guidelines in Tunisia on how social media data should be used. Ennouri stated that more effort should be done to collect as much data as possible, since data is now "like gold", and that governments should share more of their data with the public or with organizations working on AI. Data Co-Lab works with public data sets available on the internet, also scrapes data from the internet, and also uses data from applications that is has created. For Triki, open data means sharing part of your data, while respecting the governing laws. Triki does not view open data as something beneficial for the private sector, since sharing data will mean having more competitors. However, when it

[78] See https://opendata.cityofnewyork.us

comes to development, data openness can be beneficial, and thus the public sector needs to come up with solutions to incentivize private companies to open up their data.

Mahasneh stated that among Jordan's commitments with the Open Government Partnership (OGP) is having a clear policy for open data, in addition to commitments related to data quality and how public entities should release their data. Open data can be utilized to benefit civil society and the public sector.

Khooli stated that data availability democratizes AI, provides real cases for training AI, engages citizens in decision-making, and improves public service delivery. He shared the view that in Palestine, more work is required at the policy and legislative levels, in addition to work in educational and training institutions, for open data to solve developmental issues. Khooli warned that while data availability is essential for AI products and services, data quality issues may outweigh openness. For example, one should be weary of data designed to improve public image or data designed to encourage specific activities by the data providers. Official records, bearing in mind privacy concerns, provide an excellent data source for socio-economic purposes. Data can also be collected through crowd sourcing, from social media or the web, via publicly available datasets, or using the Internet of Things (IoT). Khooli personally works with Arabic text to produce deep learning language models (text generation, classification, translation and conversational AI). This work needs sizable text corpora, for which he has used Arabic Wikipedia, social media, and web crawled texts, although they are still not enough. In some cases, he had to personally machine translate English datasets.

According to Mavridis, open data is public data that anyone can use for their own benefit, in order to analyze things, to create venture, and so on. Open data shares similar characteristics with the open innovation and open software movements, whereby anyone can analyze data for their own benefit. Mavridis added that it is not a matter of having things open, but easily accessible for people to use them and understand them. However, he notes that when it comes to open data, a further difficulty, beyond privacy, is the potential for severely malignant uses of such data, even if it is anonymized appropriately. Therefore, although openness is certainly very attractive and does provide strong benefits to quicker progress, it needs to be supported and expanded in scope and volume, but it also needs careful consideration and gradual experimental deployments sometimes.

Halawi added that data openness (or lack thereof) plays a huge role in many of Lebanon's problems.

F. What capacities are necessary for (your country/ MENA) to make the best of new technologies? (human capital, infrastructure, and legislative environment)

For Egypt, Ismail believes that the main issue is not human capital, rather it is legislation and infrastructure. The legislative environment is prohibiting to new technologies. For instance, financial service legislation is limiting and data protection legislation is inadequate. Infrastructure needs more investment, including 5G and capacities for hosting large data centers. In Abaza's view Egypt has many engineers, but they lack key design thinking and conceptualization skills. Such skills are needed for AI design. In Radwan's viewpoint, Egypt needs the upskilling and reskilling of people, which requires educational programs to shift from specific knowledge to skills. Egypt, like other countries, is working on this aspect. Radwan added that this is being done through national efforts for technology literacy, general skill improvement (such as communications, team working, adapting and transferring knowledge, and creative thinking), and the collaboration of entities and stakeholders in the National Council for AI.

All the Tunisian interviewees voiced their concern with the lack of appropriate educational modules to cope with new technologies. The prevalent method of learning about AI in Tunisia is done online via self-teaching. Halioui added that there are more and more masters programs and license-level education programs launched every year in the public education system in Tunisia, but this phenomenon is a natural adaptation of the teachers and students. Haddad added that curriculums are old fashioned, and Triki explained that there are no public universities in Tunisia offering AI modules. Tunisian youth overcome this hurdle by taking online courses, and getting certifications from abroad. Brain drain is another concern mentioned by Haddad and Ennouri, due to unattractive work conditions. To make the best of new technologies, Halioui shared that more community building is needed. The AI community in Tunisia lacks access to spaces, events, and conferences to share experiences, learn, and co-create. Halioui believes that infrastructure is progressing steadily and exponentially, regardless of the issues that exist. Haddad voiced the need for a faster and more stable internet connection. Triki shared the same concern, mentioning the need for data search services and computing power capacities.

As for the legislative environment in Tunisia, both Halioui and Ennouri acknowledge that there has been support for new technologies, but there remains work to be done. Other capacities needed include access to the Internet, education in coding, and accelerating the upgrades of technical programs and software engineering, in addition to more funding for startups. Haddad explained that there are currently two directions when it comes to the legislative framework for new technologies in Tunisia. One direction is pushing for the AI revolution, with initiatives from the government such as the Startup Act (despite its slow application). The other direction is ignoring new technologies completely. As a startup owner, Haddad faces the hurdles of slow administration and bureaucracy.

For Jordan, Mahasneh explained that the system as a whole does not produce an environment conducive to the development of AI. Mahasneh explained that startups face issues with regards to laws. Jordan does not have a Startup Act like Tunisia, so starting a new business is not an easy task and you are faced with many bureaucratic issues. Internet is not an obstacle for Jordan. As for human capital, Mahasneh stated that there are differing perspectives on the matter. One point of view is that Jordan has high caliber candidates (ICT graduates and skilled engineers), and the other perspective, which is that of hiring companies, voices the struggles with finding the needed talents. Khooli from Palestine explained that there are vital pillars necessary to make the best of new technologies: capacity, infrastructure, datasets, and legal framework.

Mavridis mentioned that legislation is very important for new technologies. He added that an interdisciplinary framework with individuals from different academic backgrounds contributing to forward-looking legislation is vital. Human capital was mentioned as another important area, where also interdisciplinarity is needed. While is a need for individuals with the technical skills, there is also a need for decisionmakers, technology ethicists, legal experts, and those involved in media dissemination. Network connection is also important, added Mavridis.

Halawi stated that you cannot count of one type of capacity to make the best of new technologies, you need an alliance of academia, entrepreneurs, the private sector, in addition to investing in those projects. Both infrastructure and the legislative environment need improvement in Lebanon.

G. Should governments play a role in supporting AI and AI for development?

For Egypt, Abaza explained that legislation is an issue that the government can work to fix, through using policy to avail data so startups can train algorithms. In Ismail's view, the government should act as an enabler and fill in the gaps created by the private sector. Additionally, Egypt's ability to provide legal infrastructure to attract funds is weak. Radwan stated that Egypt's personal data protection law is in effect, and it is very important to govern relations between people as data owners and data users. There is a need for data infrastructure, meaning the ability to collect data, govern it, and store it.

There were conflicting views among the Tunisian interviewees on whether governments should play a role in supporting AI for development. Haddad shared the view that the government needs to keep up with the technology revolution and create a clear public strategy on how to use AI. Haddad specifically mentioned Kais Mejri, who works in the Tunisian government sector, and trying to help startups, who should be treated differently than regular companies. Ennouri stated that the Tunisian government is trying to support to AI field, and that without such government support AI will not survive. Triki shared that the government should interfere as an encouraging agent for the field of AI through sharing public sector data and creating academic curriculums that serve AI. However, Triki stated that the government should not interfere with personal data. Halioui, on the other hand, opposed government intervention, stating that governments tend to create bureaucratic processes and slow down the innovation cycle. He added that the government would benefit from monitoring the AI space for potential useful innovations.

For Jordan, Mahasneh believes that the government should act as a regulator, supporter, and policy facilitator for the AI sector. He mentioned suggestions such as adding AI in governmental higher education curricula, creating stronger data protection laws, and acting as a public donor. The Jordanian government recently passed a law with some tax breaks for IT companies, which is very conducive to the AI environment.[79]

As for Palestine, Khooli believes that the government should be playing a role in supporting AI for development, but it is not fast enough. There is a need for legislation, policies, and national strategies for AI, in addition to funding. Khooli added that currently in Palestine there is a special ministerial committee for open data and an open data platform (CKAN).[80] However, creating a national strategy should not be an alternative to engaging in real efforts on the ground or restricting innovative horizons.

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[79] See https://www.almamlakatv.com/news/9763-الرزاز-إعفاءات-ضريبية-لشركات-تكنولوجيا-المعلومات
[80] See http://www.opendata.ps and https://ckan.org

Mavridis stated that the policies governments create should ideally aim to maximise the benefits of AI while minimizing harms. He added that France, China, and the United Kingdom are exemplary of government support for AI development, including creating the right conditions to incentivize private sector investment, in addition to creating centers for data ethics. Government policies should tackle the adoption and creation of AI and responding to the challenges of the economy. Mavridis gave the example of governments mitigating, through policies, the negative impact that AI could have on the future of work, employment and the job
market. They must focus on utilizing AI in government and infrastructure, and bear in mind how tech can be used to maximize inclusion.

For Lebanon, Halawi believes the government should intervene in a well-designed manner, through official and specialized policymakers. There is a need for investment in entrepreneurship, in addition to tax breaks for startups.

H. What can other actors do?

For Egypt, Radwan shared several points on what other actors can do for the AI ecosystem. She mentioned that private companies need to realize and study the value of using AI for development as a core business area, beyond the umbrella of corporate social responsibility. She added that corporations also have an ethical duty to ensure their data and models are ethical and of good quality, which entails including minorities. As for the startup community, including venture capitalists and angel investors, Radwan stated that there needs to be more investment in deep technology, which requires investors to be more educated. As for students, Radwan explained that stamina for self-education is key when it comes to AI, especially so that people are able to endure the difficult learning process. Abaza added that more critical thinking and information is needed on the impact of AI, since any algorithm will give results; analyzing these results is the issue.

Ismail shared that view that universities can incorporate applied research, focusing on computer science and mathematics, and that telecommunications companies and banks should consider launching their own data labs to develop models using their own data. Additionally, he wondered why the Egyptian Ministry of Communications and Information Technology does not have government data labs, especially since it has the Information Technology Industry Development Agency (ITIDA) and the Information and Decision Support Center (IDSC). Ismail added that each sector has its own bottleneck. For example, in agriculture robotics and drones (especially since they are not legal in Egypt) are the problem and in finance, logistics, and transportation data is the problem.

From the Tunisian perspective, Halioui stated that other actors should create a supportive an enabling environment for AI to flourish. Actors can be present at many different layers of the ecosystem, including in the areas of education, funding, and mentoring. Haddad and Ennouri mentioned that other actors need information and data sharing, in addition to government awareness. Triki further elaborated that missing data is one of the problems that AI can solve through AI power solutions. Ennouri added that other actors should encourage local talents to stay in Tunisia, which would positively impact the AI ecosystem.

For Jordan, Mahasneh pointed to the importance of reforming the education system. In both the undergraduate and graduate level education, there is a gap between the outputs of the university system and the labor market demands, thus a reassessment of the priorities of universities and the necessary funding is needed.

When asked what can other actors do in Palestine, Khooli emphasized a multi-stakeholder collaborative approach, noting that, apart from government or the public sector, stakeholders or entities that can play an active role include the following: the private sector (including entrepreneurs and startups), civil society organizations, educational and training institutions, the media, and communities of practice (including researchers and data scientists). These stakeholders need to collaborate, build capacity, create synergies, share resources, and work on advocacy to bridge gaps. Existing gaps include capacity, resources, infrastructure, datasets, legal environment, logistical restrictions, competition from developed markets, and social disparity.

For Lebanon, Halawi explained that there is a need for an open data portal, in addition to the integration of advisors, data-driven actors, and governmental actors. Investors need to care about more than merely financial revenue. Students in Lebanon with AI potential are unlikely to stay in the country, since they do not believe any possibilities are available to them. There is a general lack of awareness on the potential of data and AI for the population, thus increased awareness is necessary. Other actors can also try to solve Lebanon's transportation issues through AI, yet this has not materialized since people opt to sell their technologies abroad.

VII. Conclusion and Recommendations

The state of AI is not uniform across the MENA region. In general MENA countries are mostly adaptors of existing AI technologies. Yet the dichotomy of producer vs. adaptor is less useful than perceived to be at the onset of this research. One does not supersede the other, or rather the dichotomy need not be the focal point. Rather, the extent to which a country successfully adopts AI (or other) technologies to address its economic and developmental problems is of more added value as a metric for assessment.

The starting point indeed should be identifying the most pressing developmental concerns (which intertwine with economic ones) and then assessing whether and how AI technologies are best suited to provide solutions. Indeed, the most popular forms of AI in the region stem from regional realities. These include, for example in Egypt, machine learning algorithms developed to provide financial inclusion tools such as credit scoring, where a significant portion of the population remains outside of the banking system. Driven by a need to train algorithms to understand Arabic, NLP is also extremely competitive in the region.

In the Global North, AI and new technologies development is spearheaded by the private sector. Across MENA, the state of AI varies in the public versus private sector. With the exception of the UAE, where the public sector actively leads AI efforts, interviewees cited a risk averse, reluctant public sector versus a more dynamic fast passed private sector. The dichotomy here is relevant given that the public sector plays a key role in several enablers of AI development, such as in data, infrastructure, and education. Necessary legislation, already in place in the Global North, is also still under development or trial in much of MENA. This also means governments have a significant role to play. As such, and as reflected in results, the way forward involves complementary efforts in the region between public and private players to complement and support each other. This entails collaborative means of doing so.

Empirical research in this study reiterates the principal role data plays within an AI ecosystem. This is especially true in the countries understudy in the MENA region, given the many obstacles with regards to data present in many of them. These include data quality, relevance and availability, infrastructure and management (including legislation). Mechanisms for sharing and openness as they pertain to data, are and will be key to the success of AI ecosystems in these countries, especially to ensure AI is harnessed for inclusive development.
Going forward, policies and clear implementation guidelines motivating the different players to collaborate for 'win-win' situations are needed. Players need to see their interests protected and to be made aware of the benefits of their collaboration. For example, encouraging those developing technologies and solutions, such as startups to work together with larger players sitting on a wealth of crucial data. Indeed, the seeds for such initiatives guide some of the region's national AI strategies, such as that of Egypt. Also, several countries understudy are signatories to the Open Government Partnership, such as Jordan, Tunisia in addition to Morocco. Egypt, Tunisia and Lebanon have data protection laws, and the UAE sector specific data provisions. These are important steps in encouraging more openness, transparency and more efficient and effective policy making. Clear guidelines for implementation of strategies, laws and policies is an area that requires further pressing attention.

Beyond legislation and initiatives, successful implementation will also require nurturing a culture of trust amongst players. Data, especially data held by large corporate players, is critical to competitiveness. Consumers are also understandably reluctant about sharing their own data, especially in the absence of effective mechanisms to protect them. Much like other regions, it is a delicate balance. More data openness is needed to avoid data monopolies, via perhaps anonymized data, while also considering privacy. Perhaps this also necessitates evolving business models whereby certain data is made available for R&D purposes, or selective openness. This study finds that the starting point for players needs to be identifying developmental problems, and the data necessary to develop AI or other technology solutions. Perhaps this is a more manageable, measurable process. A clearly pressing challenge that could unite players to develop AI solutions would be the state of healthcare in the region especially in light of the Covid-19 pandemic.

The pervasiveness of artificial intelligence and new technologies means changes and adaptations are necessary across the board: in legislation, in infrastructure, and in human capacities and skills. With regards to legislation, beyond data-centered legislation, financial services' regulation for example, also needs to be in line with the goals of AI for development. Interviewees in this study stressed the important role of domain experts in developing technologies and also the need for them to understand the benefits their sectors stand to gain on the back of incorporating new technologies. New and dynamic education modules are needed in schools, universities especially those that stress design thinking, conceptualization and data literacy. Skills training and re-training covered in the literature on AI capacities is also crucial. Grounds up efforts, including self-learning may also play a part, such as in Tunisia where online courses were an effective tool for some to master new technologies. Still, those with access to these courses were at an advantage of access to the internet and to computers. To ensure the process is more inclusive, government and civil society led efforts and collaborations are necessary.

There is potential for artificial intelligence and new technologies to be harnessed for inclusive development in MENA. The Covid-19 pandemic has highlighted this reality of the role technologies can play in human lives. The state of AI, data and MENA, with the exception of regional leaders, continues to be nascent and requires collaborative efforts amongst stakeholders for positive and sustainable progress.

- **Annex I: Interview Questions**

1. Could you briefly describe what you think the current state of AI in (country of study)/ MENA is?

1.1. What new technologies do you consider to be AI and 4IR based?

1.2. Which are the most popular (currently, or in the future?) in the MENA region or in your country? (3D printing? Robotics? Machine learning?)

1.3. Is your country/ MENA producers or adaptors of AI? does it matter?

2. Does the state of AI and new technologies differ in the public vs. private sector? How?

3. On a scale of 1 to 10, how important is data to the state of AI in your country/ MENA?

3.1. What type of data is important to the different players? (public vs. private players?)

3.2. What issues do you face in terms of data use or production?

3.3. How would you like to see issues you face with data change for your benefit/ benefit of the ecosystem?

3.4. Do you prefer more traditionally collected data? Or do engage in alternative data collection methods? Why or why not?

4. What does AI for Development mean or entail to you?

4.1. What, in your opinion are the region or your country's 3 most pressing developmental concerns?

4.2. Do new technologies have any potential role in mitigating these challenges? Specifically ask about gender issues here.

4.3. Is it something you are directly concerned with?

5. Are you familiar with the concept of data openness? (yes/no)

5.1. What does data openness mean in your understanding?

5.2. Is it important to you? (If so, how?)

5.3. What role if any can data openness play in AI for development?

5.4. What concerns if any do you have with data openness?

6. What capacities are necessary for (your country/ MENA) to make the best of new technologies?

6.1.1. How important are human capital, infrastructure, and the legislative environment?

7. Do/ should governments play a role in supporting AI for development?

7.1.1. What support would you like to see? 7.1.2. Examples of existing support? 7.1.3. Where should they not intervene? 7.1.4. Are there any policies or policy direction you think is needed to foster AI for development in the region?

8. What can other actors in the data/AI ecosystem do? What do they need to flourish? What gaps do they face?

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